

EPIC Meeting

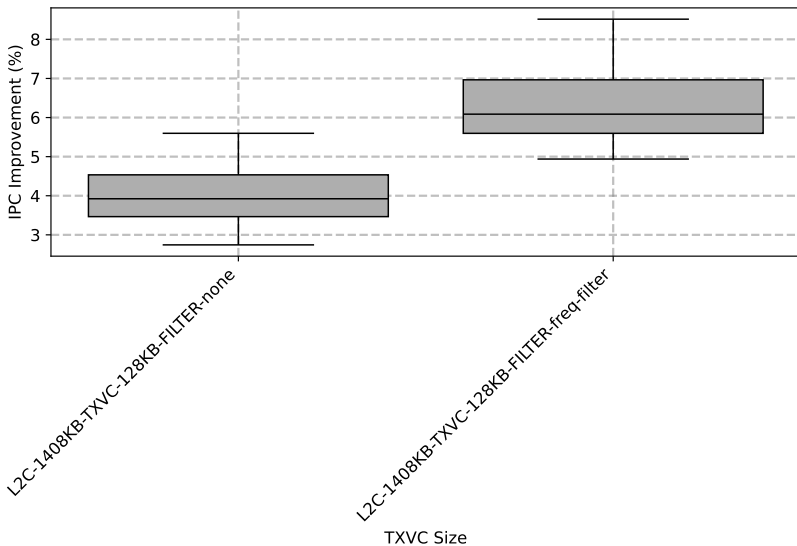
ChampSim, Set indexing Bug

Alexander V. Jamet Dimitrios Chasapis Georgios Vavouliotis
Marc Casas

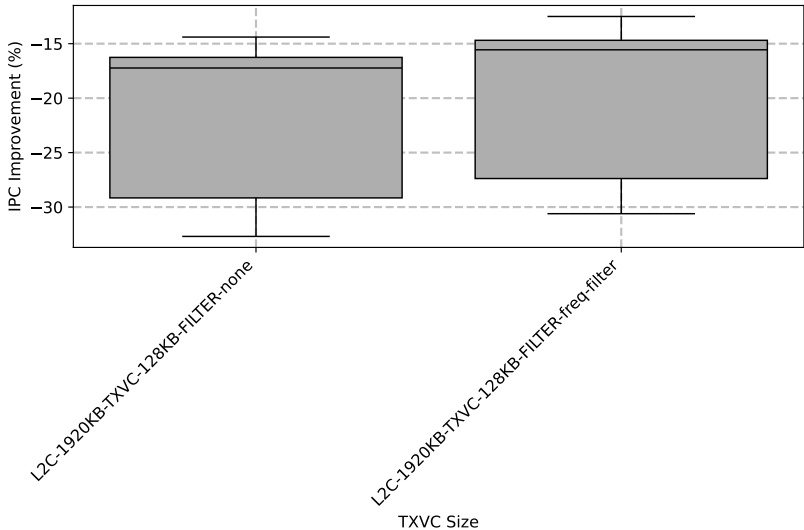
Barcelona Supercomputing Center (BSC)

Content

TXVC-128KB vs L2C 1535KB



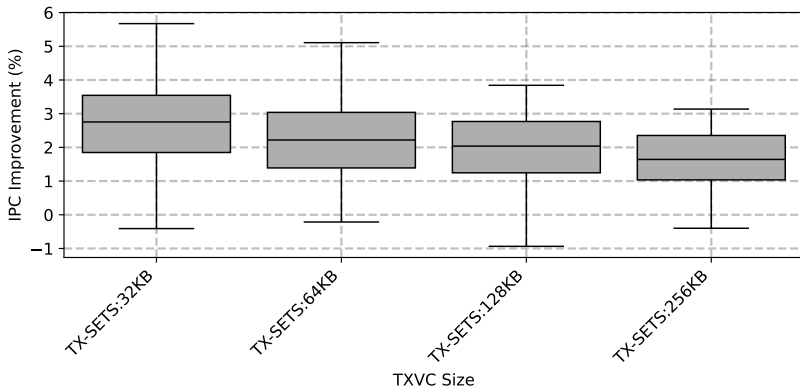
TXVC-128KB vs L2C 2048KB



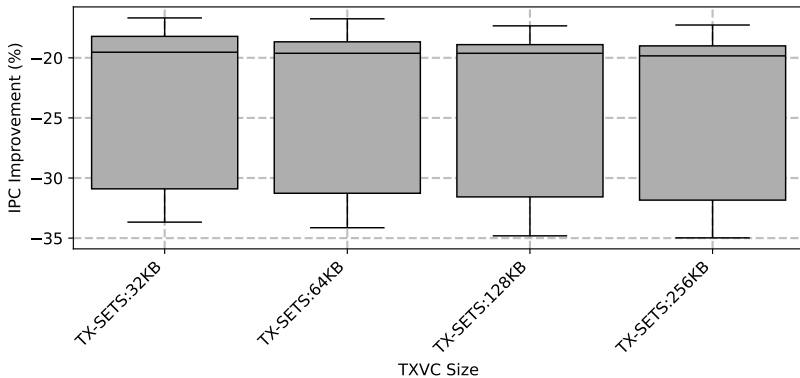
Alternative implementation using SETS

- Steal some sets of the cache to use exclusively for PTEs
- ...

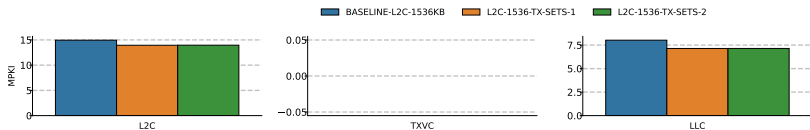
TX-SET vs L2C 1536KB



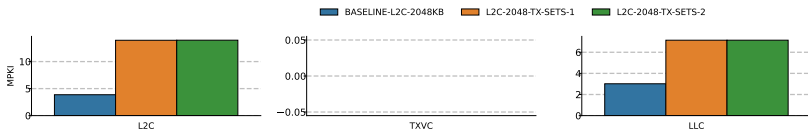
TX-SET vs L2C 2048KB



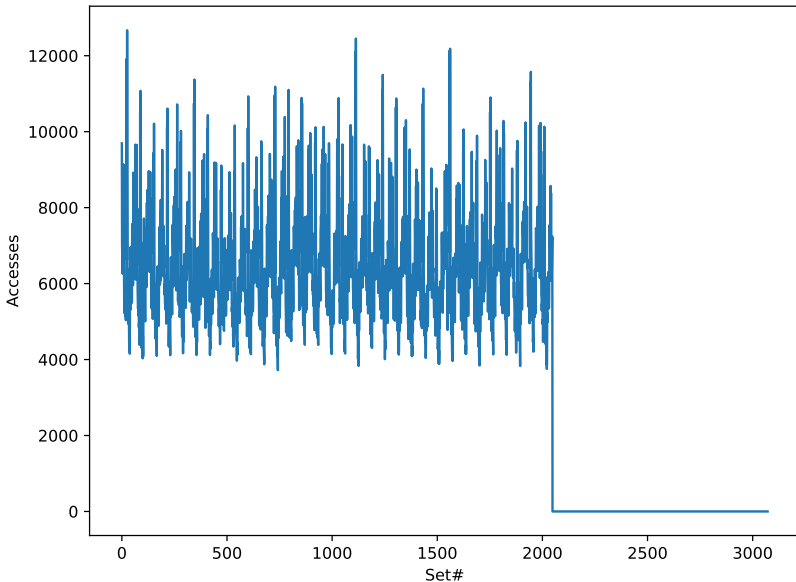
TX-SET vs L2C 1536KB, closer look -> MPKI



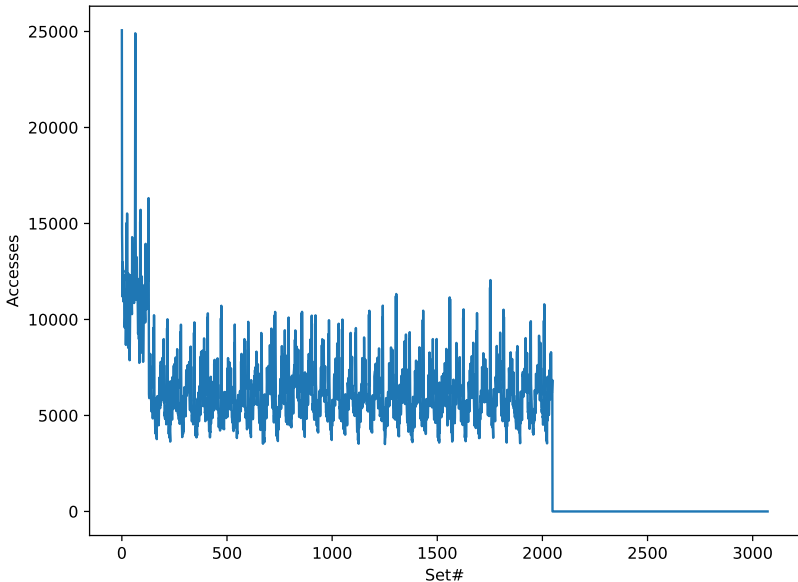
TX-SET vs L2C 2048KB, closer look -> MPKI



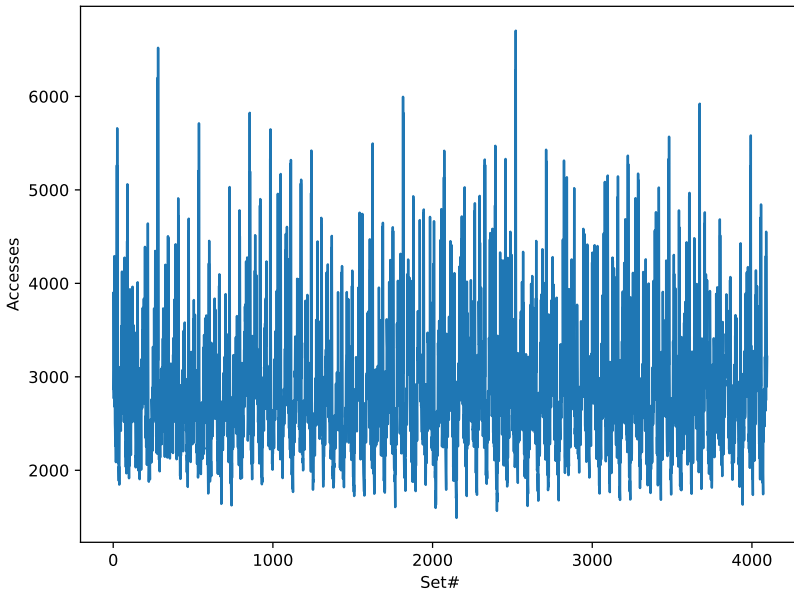
TX-SET vs L2C 1536KB, closer look -> set utilization



TX-SET vs L2C 1536KB, closer look -> set utilization



TX-SET vs L2C 2048KB, closer look -> set utilization



TX-SET vs L2C 2048KB, closer look -> set utilization

